UNIT - II

Interpolation & Curve Fitting

(Finite differences, Newton forward / backward, Lagrange, Least squares)

1. Introduction to Interpolation

Suppose values of a function $y = f(x)$ are known only at a few points $x_0, x_1, \ldots, x_n$ (a table of data). The process of finding the value of y at a point INSIDE the range of the table is INTERPOLATION; outside the range it is EXTRAPOLATION.

- *Equal spacing ($x_{i+1} - x_i = h$): use Newton forward / backward formulae.*
- *Unequal spacing: use Lagrange's formula (or divided differences).*
- *The interpolating polynomial $P_n(x)$ of degree $\le n$ passes through all $n+1$ given points and is UNIQUE.*

2. Finite Differences

Forward differences (Δ)

Let $y_0, y_1, \ldots$ be the values at $x_0,\ x_0 + h,\ x_0 + 2h, \ldots$ Then

$$\Delta y_0 = y_1 - y_0, \quad \Delta y_1 = y_2 - y_1, \quad \ldots \quad \Delta y_r = y_{r+1} - y_r$$

$$\Delta^2 y_r = \Delta y_{r+1} - \Delta y_r, \qquad \Delta^n y_r = \Delta^{n-1} y_{r+1} - \Delta^{n-1} y_r$$

Backward differences (∇)

$$\nabla y_r = y_r - y_{r-1}, \qquad \nabla^2 y_r = \nabla y_r - \nabla y_{r-1}$$

Note: $\nabla y_1 = \Delta y_0$, $\nabla^2 y_2 = \Delta^2 y_0$ - the same numbers appear in a diagonal-vs-horizontal position of the table.

Central difference (δ) and shift operator (E), averaging (μ)

$$\delta y_r = y_{r+\frac{1}{2}} - y_{r-\frac{1}{2}}, \qquad E y_r = y_{r+1}, \qquad E^n y_r = y_{r+n}$$

Operator relations

$$\Delta = E - 1, \qquad \nabla = 1 - E^{-1}, \qquad \delta = E^{1/2} - E^{-1/2}$$

$$E = 1 + \Delta, \qquad \nabla = \Delta E^{-1}, \qquad \mu = \frac{1}{2}(E^{1/2} + E^{-1/2})$$

Also: $\Delta\nabla = \nabla\Delta = \delta^2$, $E\nabla = \Delta$, $(1+\Delta)(1-\nabla) = 1$.

Important results

- Δ^n *of a polynomial of degree* n *is a CONSTANT equal to* $n!\, a_0\, h^n$; $\Delta^{n+1} = 0$.
- $\Delta(c) = 0$, $\Delta(af + bg) = a\Delta f + b\Delta g$ *(linear operator).*
- $\Delta^n y_0 = \sum_{r=0}^{n} (-1)^r \binom{n}{r} y_{n-r}$, *e.g.* $\Delta^2 y_0 = y_2 - 2y_1 + y_0$, $\Delta^3 y_0 = y_3 - 3y_2 + 3y_1 - y_0$.
- $y_n = E^n y_0 = (1+\Delta)^n y_0 = y_0 + n\Delta y_0 + \binom{n}{2}\Delta^2 y_0 + \ldots$

Example 1. Forward difference table for $y = x^3 + x$ (x = 0,1,2,3,4,5)

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
0	0	2	6	6	0
1	2	8	12	6	0
2	10	20	18	6	
3	30	38	24		
4	68	62			

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
5	130				

$\Delta^3 y = 6 = constant = 3! \times 1 \times 1^3$ (degree 3, $a_0 = 1$), and $\Delta^4 y = 0$.

3. Newton's Forward Interpolation Formula

Used when the point lies NEAR THE BEGINNING of the table.

Take $x = x_0 + uh$, *i.e.* $u = \dfrac{x - x_0}{h}$.

Newton-Gregory forward formula

$$y(x) = y_0 + u\,\Delta y_0 + \frac{u(u-1)}{2!}\Delta^2 y_0 + \frac{u(u-1)(u-2)}{3!}\Delta^3 y_0 + \cdots$$

Terms continue until the differences vanish / table ends.

Derivation (sketch): since $y_u = E^u y_0 = (1+\Delta)^u y_0$, expand by the binomial theorem: $(1+\Delta)^u = 1 + u\Delta + \frac{u(u-1)}{2!}\Delta^2 + \cdots$

Example 2. Population data: find the population in 1895

Year x	**Pop.** y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
1891	46	20	-5	2	-3
1901	66	15	-3	-1	
1911	81	12	-4		
1921	93	8			
1931	101				

$$x = 1895,\ x_0 = 1891,\ h = 10 \ \Rightarrow\ u = \frac{1895 - 1891}{10} = 0.4$$

$$y(1895) = 46 + (0.4)(20) + \frac{(0.4)(-0.6)}{2}(-5) + \frac{(0.4)(-0.6)(-1.6)}{6}(2)$$

$$+ \frac{(0.4)(-0.6)(-1.6)(-2.6)}{24}(-3)$$

$$= 46 + 8 + 0.60 + 0.128 + 0.1248 \Rightarrow y(1895) \approx 54.8528 \text{ (about } 54.85$$

thousands).

Example 3. Given sin 45 = 0.7071, sin 50 = 0.7660, sin 55 = 0.8192, sin 60 = 0.8660; find sin 52

x°	$y = \sin x$	Δy	$\Delta^2 y$	$\Delta^3 y$
45	0.7071	0.0589	-0.0057	-0.0007
50	0.7660	0.0532	-0.0064	
55	0.8192	0.0468		
60	0.8660			

$u = \dfrac{52-45}{5} = 1.4.$

$\sin 52^\circ = 0.7071 + 1.4(0.0589) + \dfrac{1.4(0.4)}{2}(-0.0057) + \dfrac{1.4(0.4)(-0.6)}{6}(-0.000$

$\Rightarrow \sin 52^\circ \approx 0.7880$ *(table value 0.7880).*

4. Newton's Backward Interpolation Formula

Used when the point lies NEAR THE END of the table. Take the last point x_n *and* $v = \dfrac{x - x_n}{h}$ *(negative).*

Newton-Gregory backward formula

$$y(x) = y_n + v\,\nabla y_n + \frac{v(v+1)}{2!}\nabla^2 y_n + \frac{v(v+1)(v+2)}{3!}\nabla^3 y_n + \cdots$$

(Derived from $y_v = E^v y_n = (1-\nabla)^{-v} y_n$*.) Differences used are those along the LAST ROW (diagonal) of the table.*

Example 4. Using the data of Example 3 find sin 57

$v = \dfrac{57-60}{5} = -0.6$; $\nabla y_n = 0.0468$, $\nabla^2 y_n = -0.0064$, $\nabla^3 y_n = -0.0007$

(last entries of each column).

$\sin 57^\circ = 0.8660 + (-0.6)(0.0468) + \dfrac{(-0.6)(0.4)}{2}(-0.0064) + \dfrac{(-0.6)(0.4)(1.4}{6}$

(table 0.8387).

Example 5. Same population table: find population in 1925

$v = \dfrac{1925 - 1931}{10} = -0.6;$

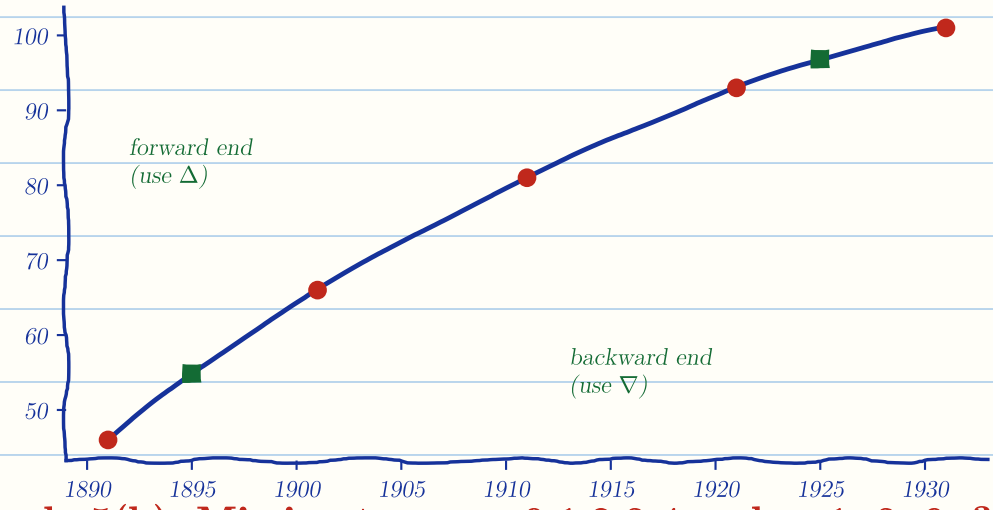
$\nabla y_n = 8,\ \nabla^2 y_n = -4,\ \nabla^3 y_n = -1,\ \nabla^4 y_n = -3.$ *Then*

$$y(1925) = 101 + (-0.6)(8) + \frac{(-0.6)(0.4)}{2}(-4) + \frac{(-0.6)(0.4)(1.4)}{6}(-1)$$

$$+ \frac{(-0.6)(0.4)(1.4)(2.4)}{24}(-3)$$

$y(1925) \approx 96.8368$ *(about 96.84 thousands).*

Forward formula near start, backward near end



Example 5(b). Missing term: x: 0,1,2,3,4 and y: 1, 3, 9, ?, 81 - find the missing value

Four values are known, so the polynomial has degree ≤ 3 *and hence* $\Delta^4 y_0 = 0$*:*

$$y_4 - 4y_3 + 6y_2 - 4y_1 + y_0 = 0 \Rightarrow 81 - 4y_3 + 54 - 12 + 1 = 0 \Rightarrow y_3 = 31$$

Rule: with k *known values,* $\Delta^k y = 0$ *(assume polynomial of degree* $k-1$*).*

Error in interpolation

If f *is* $(n+1)$ *times differentiable, the error of the* n*-th degree interpolating polynomial is*

$$E(x) = f(x) - P_n(x) = \frac{(x-x_0)(x-x_1)\cdots(x-x_n)}{(n+1)!} f^{(n+1)}(\xi), \quad \xi \in (x_0, x_n)$$

The error is smallest near the middle of the table and largest near the ends; extrapolation is therefore unreliable.

5. Lagrange's Interpolation Formula (unequal intervals)

Given $(x_0, y_0), (x_1, y_1), \ldots, (x_n, y_n)$ with ARBITRARY spacing, the polynomial of degree n through them is

Lagrange's formula

$$y = f(x) = \sum_{i=0}^{n} y_i \prod \frac{x - x_j}{x_i - x_j}$$

$$= \frac{(x-x_1)(x-x_2)\cdots(x-x_n)}{(x_0-x_1)(x_0-x_2)\cdots(x_0-x_n)} y_0 + \frac{(x-x_0)(x-x_2)\cdots(x-x_n)}{(x_1-x_0)(x_1-x_2)\cdots(x_1-x_n)} y_1 + \cdots$$

- *For 3 points (x_0, x_1, x_2) there are three terms; each term omits its own factor in numerator & denominator.*
- *INVERSE interpolation: interchange the roles of x and y in the formula to find x for a given y.*
- *No need for equal spacing and no difference table; but adding a new point means recomputing everything.*

Example 6. Given (5,12), (6,13), (9,14), (11,16) find $y(10)$

At $x = 10$: numerator $\prod(x - x_j)$ and denominator $\prod(x_i - x_j)$ (over $j \neq i$):

Point	Numerator	Denominator	y_i	Term
x_0=5	-4	-24	12	2.0000
x_1=6	-5	15	13	-4.3333
x_2=9	-20	-24	14	11.6667
x_3=11	20	60	16	5.3333

$y(10) = 14.6667 \approx 14.6667$.

Example 6(b). Inverse interpolation: for the same data find x when $y = 15$

Interchange $x \leftrightarrow y$: $x = \sum x_i \prod_{j \neq i} \frac{y - y_j}{y_i - y_j}$, $y = 15$.

Weights $L_0 = 0.2500,\ L_1 = -1.0000,\ L_2 = 1.5000,\ L_3 = 0.2500 \Rightarrow$ $x \approx 11.5000$.

Example 7. Find the polynomial through (0,2), (1,3), (2,12), (5,147)

Using Lagrange's formula and simplifying term by term (or by differences), we get

$$f(x) = x^3 + x^2 - x + 2$$

Check: $f(1) = 1 + 1 - 1 + 2 = 3$; $f(2) = 8 + 4 - 2 + 2 = 12$; $f(5) = 125 + 25 - 5 + 2 = 147$. *Also* $f(3) = 27 + 9 - 3 + 2 = 35$.

Verified numerically: coefficients = 1.0, 1.0, -1.0, 2.0

6. Curve Fitting - Method of Least Squares

Given n data points (x_i, y_i) we want the curve $y = f(x)$ that fits the trend. Let $e_i = y_i - f(x_i)$ be the residuals. The LEAST SQUARES method chooses the constants to minimise

$$E = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} [y_i - f(x_i)]^2$$

Setting $\frac{\partial E}{\partial a} = 0$, $\frac{\partial E}{\partial b} = 0$, … *gives the NORMAL EQUATIONS.*

(a) Straight line $y = a + bx$

Normal equations (2 unknowns a, b)

$$\sum y = na + b\sum x$$
$$\sum xy = a\sum x + b\sum x^2$$

Solving: $b = \dfrac{n\sum xy - \sum x \sum y}{n\sum x^2 - (\sum x)^2}$, $a = \bar{y} - b\bar{x}$.

Example 8. Fit a straight line to x: 1,2,3,4,5 and y: 14,27,40,55,68

x	y	xy	x^2
1	14	14	1
2	27	54	4
3	40	120	9
4	55	220	16
5	68	340	25
15	204	748	55

Last row = sums: $\sum x = 15$, $\sum y = 204$, $\sum xy = 748$, $\sum x^2 = 55$,

$n = 5$.

$204 = 5a + 15b$, $748 = 15a + 55b \Rightarrow b = 13.6$, $a = -0.0$.

$$y = -0.0 + 13.6\,x$$

(b) Second-degree parabola $y = a + bx + cx^2$

Normal equations (3 unknowns a, b, c)

$$\sum y = na + b\sum x + c\sum x^2$$
$$\sum xy = a\sum x + b\sum x^2 + c\sum x^3$$
$$\sum x^2 y = a\sum x^2 + b\sum x^3 + c\sum x^4$$

SHORT-CUT: if x values are equally spaced, shift the origin ($u = (x - \text{mid})/h$) so that $\sum u = \sum u^3 = 0$; the equations simplify greatly.

Example 9. Fit $y = a + bx + cx^2$ to x: 0,1,2,3,4 and y: 1, 1.8, 1.3, 2.5, 6.3

x	y	x^2	x^3	x^4	xy	x^2y
0	1	0	0	0	0	0
1	1.8	1	1	1	1.8	1.8
2	1.3	4	8	16	2.6	5.2
3	2.5	9	27	81	7.5	22.5
4	6.3	16	64	256	25.2	100.8
10	12.9	30	100	354	37.1	130.3

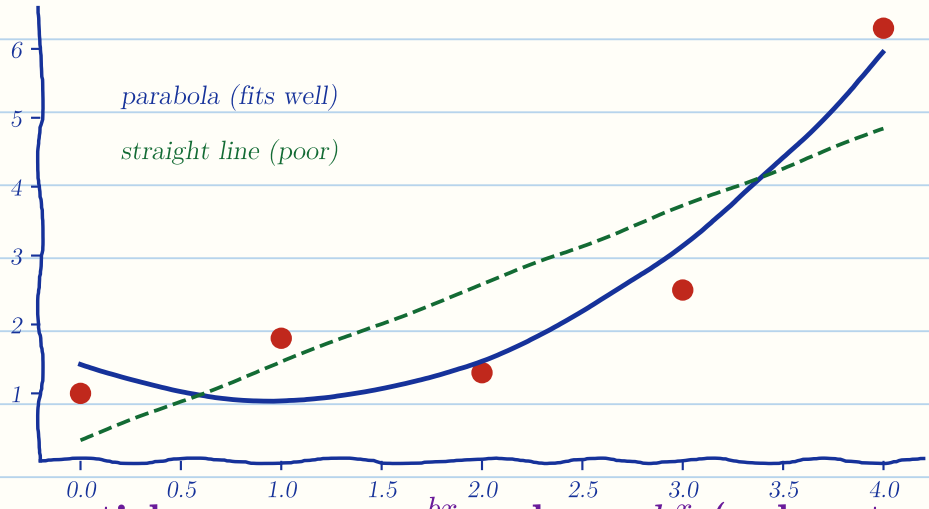
$12.9 = 5a + 10b + 30c$; $37.1 = 10a + 30b + 100c$;

$130.3 = 30a + 100b + 354c$.

Solving: $a = 1.420,\ b = -1.070,\ c = 0.550.$

$$y = 1.42 - 1.07\,x + 0.55\,x^2$$

Least-squares fits to Example 9



(c) Exponential curves $y = ae^{bx}$ and $y = ab^x$ (reduce to a line)

Take natural log: $y = ae^{bx} \Rightarrow \ln y = \ln a + bx$. *Put* $Y = \ln y$, $A = \ln a$: *line* $Y = A + bx$. *Normal equations:*

$$\sum Y = nA + b\sum x, \qquad \sum xY = A\sum x + b\sum x^2$$

Then $a = e^A$. *For* $y = ab^x$: $\log y = \log a + x\log b$ (*use* $\log_{10}$, *then* $a = 10^A$, $b = 10^B$). *For power curve* $y = ax^b$: $\log y = \log a + b\log x$.

Example 10. Fit $y = ae^{bx}$ to x: 1,2,3,4,5,6 ; y: 7.209, 5.265, 3.846, 2.809, 2.052, 1.499

x	y	$Y = \ln y$	xY	x^2
1	7.209	1.9753	1.9753	1
2	5.265	1.6611	3.3222	4
3	3.846	1.3470	4.0411	9
4	2.809	1.0328	4.1313	16
5	2.052	0.7188	3.5941	25
6	1.499	0.4048	2.4288	36
21		7.1399	19.4928	91

$7.1399 = 6A + 21b$, $19.4928 = 21A + 91b \Rightarrow A = 2.2893$, $b = -0.3141$, $a = e^A = 9.868$.

$$y = 9.87\,e^{-0.31\,x}$$

So the data follow the law $y \approx 10\,e^{-0.3x}$.

7. Quick Revision

Formula sheet

$u = \dfrac{x - x_0}{h}$ (forward, start of table), $v = \dfrac{x - x_n}{h}$ (backward, end of table).

Forward: $y_0 + u\Delta y_0 + \dfrac{u(u-1)}{2!}\Delta^2 y_0 + \cdots$ Backward: $y_n + v\nabla y_n + \dfrac{v(v+1)}{2!}\nabla^2 y_n + \cdots$

Lagrange for unequal spacing; sum of terms $y_i \prod \dfrac{x - x_j}{x_i - x_j}$.

Least squares: write the table of $\sum x, \sum y, \sum xy, \sum x^2$ FIRST, then the normal equations.

Exponential / power law: take logs, fit a straight line, then convert back.

- In exams always draw the difference table neatly and show u (or v) before substituting.
- Interpolating polynomial is unique, so Newton and Lagrange give the same answer for the same points.

8. Practice Problems (with answers)

1. Given $f(1) = 1,\ f(2) = 8,\ f(3) = 27,\ f(4) = 64$ find $f(2.5)$ by Newton forward formula. Ans: 15.625
2. Find $f(2)$ from $f(0) = 1,\ f(1) = 3,\ f(3) = 13$ using Lagrange. Ans: 7 (polynomial $x^2 + x + 1$)
3. Fit a line to (0,1), (1,3), (2,2), (3,4). Ans: $y = 1.3 + 0.8x$
4. Show that $\Delta^2 y_0 = y_2 - 2y_1 + y_0$ and $E = 1 + \Delta$.
5. Fit $y = ab^x$ to the data x: 2,3,4,5,6 ; y: 144, 172.8, 207.4, 248.8,

298.5. *Hint: use* $\log y = \log a + x \log b$. *Ans:* $y \approx 100\,(1.2)^x$